8. Develop an Android application implementing Notifications using Notification Manager.

Project Name: NotificationDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:gravity="center">

   <Button

       android:id="@+id/btnNotify"

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Show Notification"/>

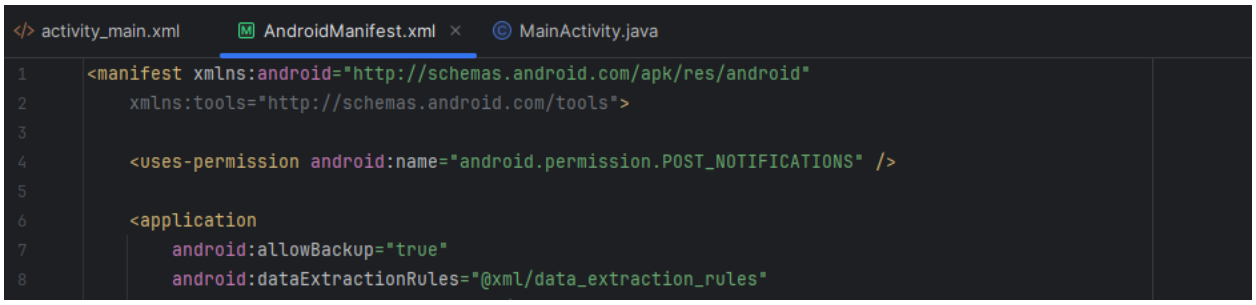
</LinearLayout>
```

Add Permission in Manifest,

Open: AndroidManifest.xml

Add above <application>:

<uses-permission android:name="android.permission.POST_NOTIFICATIONS”/>



File Name: MainActivity.java

```
package com.example.notificationdemo;


import android.Manifest;

import android.app.NotificationChannel;

import android.app.NotificationManager;

import android.content.pm.PackageManager;

import android.os.Build;

import android.os.Bundle;

import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.app.NotificationCompat;


public class MainActivity extends AppCompatActivity {


   private static final String CHANNEL_ID = "MyChannel";

   private static final int NOTIFICATION_PERMISSION_CODE = 100;


   private Button btnNotify;


   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);
```

```
        // Find the button from activity_main.xml
        btnNotify = findViewById(R.id.btnNotify);

        // Create notification channel
        createNotificationChannel();

        // Request notification permission for Android 13+
        requestNotificationPermission();

        // Button click
        btnNotify.setOnClickListener(v -> {
            showNotification();
        });
    }

    // Request notification permission
    private void requestNotificationPermission() {

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {

            if (checkSelfPermission(Manifest.permission.POST_NOTIFICATIONS)
                    != PackageManager.PERMISSION_GRANTED) {

                requestPermissions(
                        new String[]{Manifest.permission.POST_NOTIFICATIONS},
                        NOTIFICATION_PERMISSION_CODE
```

```
            );
        }
    }
}

// Create notification channel
private void createNotificationChannel() {

    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {

        NotificationChannel channel = new NotificationChannel(
                CHANNEL_ID,
                "My Channel",
                NotificationManager.IMPORTANCE_DEFAULT
        );

        channel.setDescription("Notification Channel for Demo");

        NotificationManager manager =
                getSystemService(NotificationManager.class);

        if (manager != null) {
            manager.createNotificationChannel(channel);
        }
    }
}
```

```
// Show notification
private void showNotification() {

    // Check permission for Android 13+
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {

        if (checkSelfPermission(Manifest.permission.POST_NOTIFICATIONS)
                != PackageManager.PERMISSION_GRANTED) {

            requestNotificationPermission();
            return;
        }
    }

    NotificationCompat.Builder builder =
            new NotificationCompat.Builder(this, CHANNEL_ID)
                    .setSmallIcon(android.R.drawable.ic_dialog_info)
                    .setContentTitle("Notification Example")
                    .setContentText("Hello from Android Notification")
                    .setPriority(NotificationCompat.PRIORITY_DEFAULT)
                    .setAutoCancel(true);

    NotificationManager manager =
            (NotificationManager) getSystemService(NOTIFICATION_SERVICE);
```

```
        if (manager != null) {

            manager.notify(1, builder.build());

        }

    }

}
```

